

Analysis Tutorial 5

Problem 13

(a) (i) Since $\Omega_2 \in \mathcal{A}_2$, it follows that $T^{-1}(\Omega_2) = \Omega_1 \in T^{-1}(\mathcal{A}_2)$

Now let $T^{-1}(A) \in T^{-1}(\mathcal{A}_2)$.

Then $A \in \mathcal{A}_2 \Rightarrow \bar{A} \in \mathcal{A}_2$

Then $T^{-1}(\bar{A}) = T^{-1}(A) \in T^{-1}(\mathcal{A}_2)$

Now suppose $\{T^{-1}(A_n)\}_{n \in \mathbb{N}} \subseteq T^{-1}(\mathcal{A}_2)$.

Then $\bigcup_{n \in \mathbb{N}} T^{-1}(A_n) = T^{-1}\left(\underbrace{\bigcup_{n \in \mathbb{N}} A_n}_{\in \mathcal{A}_2}\right) \in T^{-1}(\mathcal{A}_2)$

So $T^{-1}(\mathcal{A}_2)$ is a σ -algebra

(ii) T is $(\tilde{\mathcal{A}}_1, \mathcal{A}_2)$ -measurable if and only if $\forall A' \in \mathcal{A}_2, T^{-1}(A') \in \tilde{\mathcal{A}}_1$.

But this is the same as $T^{-1}(\mathcal{A}_2) \subseteq \tilde{\mathcal{A}}_1$.

(b) (i) $\Omega_2 \in \mathcal{A}_2$ and $T^{-1}(\Omega_2) = \Omega_1 \in \mathcal{A}_1$

So $\Omega_2 \in \mathcal{A}_2$

Now let $A \in \mathcal{A}_2$. Then $T^{-1}(A) \in \mathcal{A}_1$.

But $\overline{T^{-1}(A)} \in \mathcal{A}_1$ and $\overline{T^{-1}(A)} = T^{-1}(\bar{A})$

so $\bar{A} \in \mathcal{A}_2$

Now let $\{A_n\}_{n \in \mathbb{N}} \subseteq \mathcal{A}_2$. Then $T^{-1}(A_n) \in \mathcal{A}_1$

for any $n \in \mathbb{N}$. But $\bigcup_{n \in \mathbb{N}} T^{-1}(A_n) \in \mathcal{A}_1$ so

$T^{-1}\left(\bigcup_{n \in \mathbb{N}} A_n\right) = \bigcup_{n \in \mathbb{N}} T^{-1}(A_n) \in \mathcal{A}_1$ so $\bigcup_{n \in \mathbb{N}} A_n \in \mathcal{A}_2$

So $\mathcal{A}_2$ is a σ -algebra

(ii) T is $(\mathcal{A}_1, \tilde{\mathcal{A}}_2)$ -measurable

$\Leftrightarrow T^{-1}(\tilde{\mathcal{A}}_2) \subseteq \mathcal{A}_1$.

Let $A \in \tilde{\mathcal{A}}_2$. Then $T^{-1}(A) \in \mathcal{A}_1$ and $T^{-1}(A) \in T^{-1}(\mathcal{A}_2) \in \mathcal{A}_1$.

But $A \subseteq \Omega_2 \Rightarrow A \in \mathcal{A}_2$ so $\tilde{\mathcal{A}}_2 \subseteq \mathcal{A}_2$

Problem 14

(a) We need that $T^{-1}(\mathcal{P}(\Omega_2)) \subseteq \{\emptyset, \Omega_1\}$

Let $\omega \in \Omega_1$. Then $T(\omega) \in \Omega_2$.

But then $T^{-1}(\{T(\omega)\}) = \Omega_1$ (as $\neq \emptyset$),

so for every $\omega' \in \Omega_1$, $T(\omega') = T(\omega)$.

The other implication is obvious,

as for $A \subseteq \mathcal{P}(\Omega_2)$, if $T(\omega) = c$, then

$T^{-1}(A) = \Omega_1$ if $c \in A$, and $T^{-1}(A) = \emptyset$ if $c \notin A$.

(b) We know $\mathcal{B} = \sigma(\{(-\infty, x] : x \in \mathbb{R}\})$

So clearly for $x \in \mathbb{R}$, $(-\infty, x] \in \mathcal{B}$ and so

$[-x, x] \in \mathcal{B}$ so $\sigma(A) \subseteq \mathcal{B}$. Let $\tilde{\mathcal{B}} \in \mathcal{B}$.

If $\tilde{\mathcal{B}} \in \mathcal{B}$, then $\mathcal{B} \in \sigma(A)$ trivially.

If $\mathcal{B} = X \cup \{\infty\}$, $X \in \mathcal{B}$, then since

$\{\infty\} = \bigcap_{n \in \mathbb{N}} [-n, \infty] = \bigcap_{n \in \mathbb{N}} (n, \infty]$, $\mathcal{B} \in \sigma(A)$. Also,

$\{-\infty\} = \bigcap_{n \in \mathbb{N}} [-\infty, -n]$ and so if $\mathcal{B} = X \cup \{-\infty\}$

or $\mathcal{B} = X \cup \{-\infty, \infty\}$, $\mathcal{B} \in \sigma(A)$. So $\mathcal{B} = \sigma(A)$.

(c) (i) Suppose that $\forall x \in \mathbb{R} : \{f \leq x\} \in \mathcal{A}$.

$\Leftrightarrow \{f \leq x\} = \{f > x\}^c \in \mathcal{A}$.

Now $\{y < f \leq x\} = \{f \leq x\} \cap \{f > y\}$

So $\{f = x\} = \bigcap_{n \in \mathbb{N}} \{x - \frac{1}{n} < f \leq x\} \in \mathcal{A}$

Then $\{f > x\} = \{f > x\} \cup \{f = x\} \in \mathcal{A}$.

Also, $\{f > x\} \in \mathcal{A} \Leftrightarrow \{f \geq x\} = \{f < x\}^c \in \mathcal{A}$.

Now we have to show if $\{f > x\} \in \mathcal{A}$, then $\{f \leq x\} \in \mathcal{A}$.

So $\{f = x\} = \bigcap_{n \in \mathbb{N}} \{x < f < x + \frac{1}{n}\} \in \mathcal{A}$, so

$\{f > x\} = \{f > x\} \setminus \{f = x\} \in \mathcal{A}$

Problem 15

(a) We wish to show $\forall x \in \mathbb{R}: \{f(x)\} \in \mathcal{B}$

Now $\{f(x)\} = \underbrace{\{q \in \mathbb{Q}: \sin q(x)\}}_{\text{countable}} \cup \{r \in \mathbb{R} \setminus \mathbb{Q}: \cos r(x)\}$

Since $\cos x$ is continuous, it is measurable

So $\{r \in \mathbb{R} \setminus \mathbb{Q}: \cos r(x)\} \in \mathcal{B}$ and so $\{f(x)\} \in \mathcal{B}$

(b) Define $f_n(x) := \frac{f(x + \frac{1}{n}) - f(x)}{\frac{1}{n}}$

Then as $n \rightarrow \infty$, $\frac{1}{n} \rightarrow 0$ and $f_n(x) \rightarrow f'(x)$

So $f_n \rightarrow f'$. Each f_n is measurable, as f is continuous and thus measurable, so f' is measurable.